

CardioPredict – Heart Disease Risk Prediction System

A full-stack Machine Learning web application built with Flask, Scikit-Learn, and XGBoost to predict heart disease risk using patient vitals.

Project Structure:

```
cardiopredict/  
app/  
app.py  
templates/  
base.html  
index.html  
predict.html  
about.html  
static/  
data/  
heart.csv  
ml/  
preprocess.py  
train.py  
processed/  
models/  
venv/  
requirements.txt  
README.md
```

Requirements:

Python 3.9+
pip
Virtual Environment recommended

Setup Instructions:

1. Open project folder in VS Code

2. Create virtual environment:

```
python -m venv venv
```

Activate (Windows):

```
venv\Scripts\activate
```

3. Install dependencies:

```
pip install -r requirements.txt
```

If missing:

```
pip install flask pandas numpy scikit-learn xgboost joblib
```

Machine Learning Pipeline:

Preprocess dataset:
python ml/preprocess.py

Train models:
python ml/train.py

Run Web Application:
python app/app.py

Open:
http://127.0.0.1:5000

Test Case (Low Risk):

Age: 35
Sex: Male
Chest Pain: Typical Angina
Rest BP: 120
Cholesterol: 180
FBS: No
Rest ECG: Normal
Max Heart Rate: 170
Exercise Angina: No
Oldpeak: 0.0
Slope: Upsloping
Major Vessels: 0
Thalassemia: Normal

Expected Output:
Risk Level: Low or Medium
Probability: ~20%–45%

Models Used:
Random Forest Classifier
XGBoost Classifier
StandardScaler

Risk Categories:
<33% = Low
33–66% = Medium
>66% = High

Common Issues:

Model file missing:
Run preprocessing and training again.

Dataset column missing:
Ensure dataset contains required columns.

String to float error:

Handled in preprocess.py using encoding.

Feature Mapping:

age -> age

sex -> sex

chest pain -> cp

rest bp -> restbps

cholesterol -> chol

fbs -> fbs

ecg -> restecg

max hr -> thalch

exang -> exang

oldpeak -> oldpeak

slope -> slope

ca -> ca

thal -> thal

Future Improvements:

PDF report generation

Doctor dashboard

Authentication

API support

Mobile UI

Model explainability